

Validity-Aware AI: An Entropy-Bounded Architecture for Regime-Sensitive Inference

A. Entropy, Validity, and Regimes

Extending L0–L3 Regime Architecture to Knowledge-Graph Inference

Morten Magnusson

Independent Researcher

ORCID: 0009-0002-4860-5095

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Abstract

This section extends the L0–L3 regime architecture from physical systems to knowledge-graph-based AI systems. The core principle—that inference is valid only within regimes where data and dynamics remain consistent—applies equally to inference over structured information. We define entropy measures for knowledge graphs, map L0–L3 regimes to inference validity states, and establish detection algorithms for regime transitions. The framework provides a mathematical and epistemological foundation for building AI systems that know where they fail.

Keywords: entropy-bounded empiricism, regime architecture, knowledge graphs, AI safety, validity awareness, hallucination prevention

1 Introduction

The L0–L3 regime architecture establishes a foundational principle: **inference is valid only within the regime where data and dynamics remain consistent**. This principle was developed in the context of Energy-Flow Cosmology (EFC) and validated against 175 galaxy rotation curves (SPARC175), where regime-dependent model validity was demonstrated with $p < 0.0001$.

This section extends the framework to **knowledge-graph-based AI systems**, demonstrating that the same epistemological constraints apply to inference over structured information. The core insight is:

Model validity is a function of measurable entropy in the knowledge structure, not of model complexity or training volume.

2 Entropy in Knowledge Graphs

2.1 Definition

For a knowledge graph $G = (V, E)$ where V is a set of nodes (concepts, facts, entities) and E is a set of edges (relationships), we define **graph entropy** as:

$$\mathcal{H}(G) = \mathcal{H}_V + \mathcal{H}_E + \mathcal{H}_{VE} \quad (1)$$

Where:

Node entropy (uncertainty in concept definitions):

$$\mathcal{H}_V = - \sum_{v \in V} p(v) \log p(v) \quad (2)$$

Edge entropy (uncertainty in relationships):

$$\mathcal{H}_E = - \sum_{e \in E} p(e) \log p(e) \quad (3)$$

Structural entropy (uncertainty in graph topology):

$$\mathcal{H}_{VE} = - \sum_{(v,e) \in G} p(v, e) \log p(v|e) \quad (4)$$

2.2 Interpretation

- **Low entropy** ($\mathcal{H}(G) \rightarrow 0$): The graph represents a well-defined, internally consistent domain. Relationships are unambiguous. Inference is reliable.
- **High entropy** ($\mathcal{H}(G) \rightarrow 1$): The graph contains contradictions, contested relationships, or domains where multiple incompatible models coexist. Inference becomes unreliable.

Note on scaling: $\mathcal{H}(G)$ is normalized relative to domain baseline. Thresholds (θ) are calibrated per domain, not universal constants. A medical knowledge graph may have different baseline entropy than a product catalog—what matters is deviation from stable operating range, not absolute value.

2.3 Practical Measurement

In operational systems (e.g., Symbiosis architecture), entropy can be estimated through:

1. **Edge variance:** Standard deviation of confidence scores across edges of the same type
2. **Node disagreement:** Number of conflicting properties or definitions per concept
3. **Model disagreement:** Divergence between predictions from different embedded models
4. **Temporal instability:** Rate of change in graph structure over time

3 Regime Definitions for Knowledge Graphs

The L0–L3 architecture maps directly to knowledge-graph-based inference:

3.1 L0 — Latent (Configuration Space)

The graph contains **potential structure** but no active inference dynamics.

- Graph exists as schema or template
- No queries are being processed
- Relationships are defined but not traversed
- **Inference validity:** None (system inactive)

3.2 L1 — Stable Active (Predictable Inference)

The graph operates in a **low-entropy, well-defined domain**.

- Relationships are consistent and uncontested
- Historical patterns reliably predict current queries
- Linear approximations and cached inferences remain valid
- **Inference validity:** High confidence, reliable extrapolation within domain

Example: Querying a product catalog with fixed relationships (product $\rightarrow$ category $\rightarrow$ price)

3.3 L2 — Complex Active (Non-Linear Dynamics)

The graph contains **contested relationships, emerging patterns, or cross-domain tensions**.

- Multiple valid interpretations coexist
- Local sensitivity: small changes in input produce large changes in output
- Historical models begin to diverge from observed patterns
- **Inference validity**: Requires explicit uncertainty quantification; point estimates unreliable

Example: Querying scientific literature where competing theories explain the same phenomenon

3.4 L3 — Residual (Structure Without Process)

The graph contains **historical structure** but active dynamics have ceased.

- Information is frozen; no new edges or updates
- Queries return historical snapshots, not current state
- Structure preserves what was, not what is
- **Inference validity**: Historical reconstruction only; active prediction invalid

Operational mode: L3 is not “do nothing”—it is **retrieval without projection**. The system can still answer “what did we know then?” but must refuse “what should we expect now?” This read-only epistemic mode preserves value while preventing invalid extrapolation.

Example: A deprecated knowledge base from a discontinued project

4 The Regime Transition Function

4.1 L1 → L2 Transition

The critical transition for AI safety occurs when a knowledge domain **exits the stable regime**:

$$L1 \rightarrow L2 \quad \text{when} \quad \mathcal{H}(G) > \theta_{\text{critical}} \quad (5)$$

Where θ_{critical} is a domain-dependent threshold.

Phase-extended transition: This crossing is not instantaneous. The uncertainty regime grows before collapse becomes observable. A system may operate in a **transition zone** where L1 inference is degrading but L2 protocols have not yet engaged. Detecting this zone—not just the boundary—is critical for early warning.

This transition is characterized by:

1. **Edge variance exceeds baseline:** Confidence scores on relationships diverge
2. **Model disagreement emerges:** Different inference paths produce contradictory results
3. **Temporal instability increases:** Graph structure changes faster than models can adapt

4.2 Detection Algorithm

```
def detect_regime_transition(graph, query_context):
    """
    Detect whether inference should proceed in L1 or L2 mode.
    Returns regime classification and confidence.
    """
    # Measure local entropy around query
    subgraph = extract_relevant_subgraph(graph, query_context)

    H_nodes = compute_node_entropy(subgraph)
    H_edges = compute_edge_entropy(subgraph)
    H_structural = compute_structural_entropy(subgraph)

    H_total = H_nodes + H_edges + H_structural

    # Regime classification
    if H_total < THETA_L1_UPPER:
        return Regime.L1, confidence=1.0 - H_total
    elif H_total < THETA_L2_UPPER:
        return Regime.L2, confidence=estimate_l2_confidence(H_total)
    else:
        return Regime.L3_WARNING, confidence=0.0
```

Regime	System Behavior	Output Characteristics
L1	Standard inference	Point estimates, high confidence
L2	Uncertainty-aware	Probability distributions, explicit caveats
L2→L3	Inference degradation	Refusal to generate point estimates
L3	Historical retrieval only	No predictive claims, archival mode

Table 1: Regime-dependent system behavior

4.3 System Response by Regime

5 Why This Matters for AI

5.1 The Problem with Current Systems

Standard LLMs and RAG systems are **epistemically blind**:

- They generate outputs regardless of domain entropy
- They compensate for uncertainty with **more text, more coherence, more confident rhetoric**
- They have no mechanism to detect regime boundaries
- They cannot degrade their own epistemic status

This is the operational definition of **hallucination**: generating high-confidence outputs in high-entropy domains.

5.2 The EBE Solution

Entropy-Bounded Empiricism provides a structural alternative:

1. **Pre-inference entropy measurement**: Before generating any output, measure $\mathcal{H}(G)$ in the relevant subgraph
2. **Regime classification**: Determine whether the query falls in L1, L2, or L3
3. **Response modulation**: Adjust output format, confidence, and caveats based on regime
4. **Collapse detection**: Refuse to generate point estimates when approaching L2→L3 boundary

5.3 The Core Claim

A system that does not know where it fails is more dangerous than a system that fails often.

The goal is not to eliminate failure. The goal is to make failure **detectable and predictable**.

6 Relation to Existing Work

6.1 Builds On

- **L0–L3 Regime Architecture** (Magnusson 2026): Foundational regime definitions and transition logic
- **Symbiosis Architecture** (Magnusson 2025): Graph-vector hybrid memory for long-horizon reasoning
- **SPARC175 Validation** (EBE-SPARC175): Statistical demonstration of regime-dependent validity

6.2 Extends Beyond

This section introduces:

1. **Formal entropy definitions** for knowledge graphs (not physical systems)
2. **Detection algorithm** for regime transitions in inference
3. **System response protocol** linking entropy to output behavior

6.3 Does Not Claim

- Universal applicability without domain calibration
- Automatic threshold determination (θ_{critical} requires empirical tuning)
- Replacement of domain expertise (EBE constrains inference, does not replace understanding)

7 Summary

Section A establishes the **mathematical and epistemological foundation** for regime-aware AI:

1. **Entropy in knowledge graphs** is measurable through node, edge, and structural components
2. **L0–L3 regimes** map directly to inference validity in AI systems
3. **Regime transitions** are detectable through entropy thresholds and model disagreement
4. **System response** must change based on regime—this is not optional, it is epistemically required

The following sections (B, C) show how this foundation translates into concrete architecture (Regime-Aware RAG) and broader implications (Beyond the Black Box).

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